



M4: Invoice Leakage Requirements

Scope confirmed in session: MVP is reactive (pattern/history-based flagging after the fact), not proactive stop-payment before an invoice is paid. Focus is the 83% PO-backed invoice flow; non-PO invoices (~17-20%) are an open edge case. Risk classification must use pre-defined, transparent business rules - not an AI-determined black box.

Epic 1: Invoice-to-Contract Rate/Quantity Compliance

Description: Flag invoices where the vendor charged more than the contracted rate or billed for more units than were ordered.

Backed by: Invoice, Contract, ComplianceException, invoice_exception_value, gold_invoice_exception

User Stories

- As a Sourcing Associate, I want invoices flagged when the charged rate exceeds the contracted rate, so I can pursue a correction or credit with the vendor.
- As a Sourcing Associate, I want the dollar value of each rate discrepancy shown, so I can prioritize which discrepancies to chase first.
- As a Sourcing Associate, I want invoices flagged when freight, fuel, or accessorial fees are added that weren't part of the original PO or contract, so hidden charges don't go unnoticed.
- As a Sourcing Associate, I want invoices flagged when the invoiced quantity exceeds what was ordered on the PO, so I catch over-billing on units separately from rate discrepancies.
- As a Data/AI Engineer, I want this check scoped to the PO-backed invoice flow for MVP (83% of volume), with non-PO invoices explicitly out of scope until the matching approach is defined.
- ***In harmony with what Coupa is doing***

Epic 2: Contract Maximum Value Exposure and Forecast

Description: Flag when invoiced spend against a contract is approaching or projected to exceed the contract's maximum value.

Backed by: Contract (max value field), ContractCoverage, SpendFact, po_spend, invoice_spend

User Stories

- As a Sourcing Associate, I want to see when cumulative invoiced spend against a contract is approaching its maximum value, so I can act before the ceiling is hit.
- As a Sourcing Associate, I want a forward projection based on run-rate (e.g. spend pace over the first six months of a contract period), so I know if we're on track to exceed the max value before it happens and have time to renegotiate.
- As a Sourcing Associate, I want to see when a services contract is creeping toward its max value due to gradual scope additions (not a single large charge), so slow-building overages get caught early.

Epic 3: Contract Expiration and Auto-Renewal Alerting

Description: Surface upcoming contract expirations, especially those with automatic renewal clauses, sorted by category, so sourcing teams don't miss a renegotiation window.

Backed by: `contracts_v1` (Bronze - expiration and renewal fields already in base contract table data, no OCR required), `Contract`

User Stories

- As a Sourcing Associate, I want to see contracts in my category approaching expiration, so I have visibility before a renewal window closes.
- As a Sourcing Associate, I want contracts with automatic renewal clauses flagged distinctly from those that simply expire, so I know which ones require action to avoid an unfavorable auto-renewal (e.g. a rate increase).
- As a Sourcing Associate, I want to see when an auto-renewal is about to trigger a rate increase clause, so I'm not caught off guard after the renewal window closes.
- As a COE member, I want an aggregate view of expiring/auto-renewing contracts across all categories, so I can support governance oversight beyond a single category.

Epic 4: Contract Term and SLA Breach Detection

Description: Use OCR-extracted contract terms (e.g. SLAs) compared against operational data to detect breaches, such as repeated late deliveries, that put a contract at risk or entitle Kroger to remedies.

Backed by: OCR-extracted contract terms as a part of `M2: Document Digitization Requirements`, `ComplianceException`, operational/delivery data (source TBD)

User Stories

- As a Sourcing Associate, I want to see when a supplier has breached an SLA term (e.g. late shipments beyond an agreed threshold), so I can pursue renegotiation or a remedy.
- As a Sourcing Associate, I want the specific SLA term and the evidence of the breach (e.g. count of late shipments this month) shown together, so I don't have to manually cross-reference the contract language.
- As a Sourcing Associate, I want to see when an SLA breach entitles Kroger to a credit or remedy that hasn't been claimed, so we're not leaving money on the table.
- **High priority breach use cases/items (\$ amount or number of occurrences) - something to explore as we progress**

Epic 5: Transparent Risk Classification

Description: Classify flagged issues (rate overcharge, max-value exposure, expiration, SLA breach) into risk tiers (e.g. high/medium/low) using pre-defined, documented business rules rather than an opaque AI-driven score.

Backed by: ComplianceException, DataQualityIssue

User Stories

- As a Sourcing Associate, I want to see why an item was classified as high, medium, or low risk, so I can trust and act on the classification without needing to reverse-engineer it.
- As a Product/Design lead, I want risk-tier thresholds defined and documented with the business team before launch, so classification logic is agreed upon rather than AI-determined by default.
- **TPRM at Supplier Level - visibility into diverse suppliers**
- Write back and launch events in other systems
 - If find opportunity, can launch Epic project and pre-fill out forms
- **Add detail around navigation and routing to other systems (ex. Coupa)**
 - Pre-filled pieces to get started elsewhere
- **Give COE or business to adjust workflows without opening backend - access to data creating through system (e.g. timestamps)**
 - Ideally certain configurations and content management without downtime/deployments
 - Especially for workflows and orchestration

Cross-Epic Open Items

- Non-PO invoice matching approach not yet defined - PO-only for MVP, or also cross-reference RFQ responses and approved catalogs (Joe's open question).
- Insurance expiration tracking - system of record unclear; possible extension of Epic 3's scope pending confirmation with Carlos.
- Proactive stop-payment (catching mismatch before payment) explicitly out of MVP scope; revisit as future state once reactive flagging is proven out.
- Risk-tier thresholds need a working session with business stakeholders (Cody, Misty, Carlos) to define before build.